May 2016 FIFA Ranking

Number	Country	May 2016 ranking	Points
1	Argentina	1	1,523
2	Belgium	2	1,352
3	Brazil	7	1,251
4	Chile	3	1,348
5	Colombia	4	1,337
6	England	10	1,069
7	Germany	5	1,309
8	Portugal	8	1,184
9	Spain	6	1,277
10	Uruguay	9	1,158

Measurement vs. Scaling

Measurement

Assigning numbers or symbols to characteristics of objects, following pre-specified rules.

- One-to-one correspondence between numbers and characteristics
- Rules must be standardised and applied uniformly
- Rules must not change over objects or time

Scaling

Creating a continuum on which measured objects are located.

Example: an attitude scale from 1 (extremely unfavourable) to 100 (extremely favourable).

- *Measurement assigns a number to each participant*
- *scaling places participants along the continuum.*

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Scale Characteristics

01

Description

The unique labels are used to designate each value of the scale
All scales possess description

Examples:
1 Female, 2 Male
1=Strongly Agree,2= Disagree
Number of euros earned anually

02

Order

The relative sizes or positions of the descriptors.
Not all scales possess order

Examples:
Most preferred to least preferred:
Cinema , Theater, Pop Concert
Age group 35-49> Age group 20-24

03

Distance

The absolute differences between the scale descriptors are known and may be expressed in units
All scales with distance also have order

Example: A 4 person household has one person less than a 5 person household

04

Origin

That the scale has a unique or fixed beginning, or true zero point.

Example:
0 is the origin of annual household income

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Primary Scales of Measurement

01

Nominal

Numbers only label or tag objects. No amount is implied.

Counting, mode, percentages

02

Ordinal

Numbers rank objects — more or less — but not by how much.

+ Percentiles, quartiles, median

03

Interval

Equal distances represent equal differences. Zero point is arbitrary.

+ Mean, standard deviation

04

Ratio

Has all the properties of the plus a true, absolute zero point.

+ Geometric / harmonic mean

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Categorical Nominal Scale

- the numbers serve only as labels or tags for identifying and classifying objects.
- When used for identification, there is a strict one-to-one correspondence between the numbers and the objects.
- The numbers do not reflect the amount of the characteristic possessed by the objects.

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Ordinal Scale

- Is a ranking scale in which numbers are assigned to objects to indicate the relative extent to which the objects possess some characteristic.
- It is possible to determine whether an object has more or less of a characteristic than some other object, but no how much more or less.
- Any series of numbers can be assigned which preserves the ordered relationships between the objects.

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Interval Scale

- Numerically equal distances on the scale represent equal values in the characteristic being measured.
- It permits comparison of the differences between objects. The location of the zero point is not fixed.
- Both the zero point and the units of measurement are arbitrary.
- Any positive linear transformation of the form y=a+bx will preserve the properties of the scale.
- It is not meaningful to take ratios of scale values

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Ratio Scales

- It is meaningful to compute ratios of scale values.
- Only proportionate transformations of the form y=bx, where b is a positive constant, are allowed.
- All statistical techniques can be applied to ratio data.

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Comm Leisure Time Survey

	Nominal scale	Ordinal scale		Interval scale		Ratio scale
No.	Film genre	Preference rankings		Preference ratings		Amount (€) spent on cinema visits in the past three months
				1–7	11–17	
1	Action	1	10	7	17	200
2	Animated	5	53	5	15	35
3	Comedy	2	25	7	17	200
4	Drama	3	30	6	16	100
5	Factual	4	45	6	16	0
6	Fantasy	6	61	5	15	100
7	Light drama	7	79	5	15	0
8	Romance	8	82	4	14	0
9	Sci-fi	9	95	4	14	0
10	Suspense	10	115	2	12	10

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Comparative Scaling

- One of two types of scaling techniques in which there is direct comparison of stimulus objects with one another.

- The major benefit of comparison scaling is that small differences between stimulus objects can be detected

- reduce halo or carryover effects from one judgement to another

What could be a disadvantage?

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Comparative Scaling

- One of two types of scaling techniques in which there is direct comparison of stimulus objects with one another.
- The major benefit of comparison scaling is that small differences between stimulus objects can be detected
- reduce halo or carryover effects from one judgement to another
- The disadvantage is that the findings are not generalizable to other non-compared items

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Comparative Scaling Techniques

Paired Comparison

Participants choose between two objects at a time. Most widely used technique; n(n−1)/2 comparisons needed for n brands.

e.g. blind taste tests — New Coke won paired tests but failed in market, since brand image also drives choice.

Rank Order

Participants order several objects at once against a criterion. Requires only n−1 scaling decisions.

e.g. ranking preference for four film genres.

Constant Sum

Participants allocate a fixed total (e.g. 100 points) across attributes to reflect importance.

e.g. splitting 100 points across bottled-beer attributes such as taste, price and brand.

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Paired Comparisons

Instructions
We are going to present you with 10 pairs of bottled beer brands. For each pair, please indicate which of the two brands of beer in the pair you prefer.

Recording form

	Holsten	Stella Artois	Grolsch	Carlsberg	Budvar
Holsten		0	0	1	0
Stella Artois	1[a]		0	1	0
Grolsch	1	1		1	1
Carlsberg	0	0	0		0
Budvar	1	1	0	1	
Number of times preferred[b]	3	2	0	4	1

[a] 1 in a particular box means that the brand in that column was preferred over the brand in the corresponding row. 0 means that the row brand was preferred over the column brand.

[b] The number of times a brand was preferred is obtained by summing the 1s in each column

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Your Turn

	Stella Artois	Patricia	Pilsen	Patagonia	Norteña	Estrella Galicia
Stella Artois						
Patricia						
Pilsen						
Patagonia						
Norteña						
Estrella Galicia						
Times preferred						

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Transitivity of Preference

- implies that if Brand A is preferred to B, and Brand B is preferred to C, then Brand A is preferred to C.

- This can be violated by participants

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Rank-Order Example

Instructions

Rank the listed film genres in order of preference. Begin by picking out the genre that you like the most and assign it a number 1. Then find the second-most-preferred genre and assign it a number 2. Continue this procedure until you have ranked all the genres in order of preference. The least-preferred genre should be assigned a rank of 10.

No two genres should receive the same rank number

The criterion of preference is entirely up to you. There is no right or wrong answer. Just try to be consistent

Genre	Rank order
Action	
Animated	
Comedy	
Drama	
Factual	
Fantasy	
Light drama	
Romance	
Sci-fi	
Suspense	

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Constant Sum Scale

Instructions

Below are eight attributes of bottled beers. Please allocate 100 points among the attributes so that your allocation reflects the relative importance you attach to each attribute. The more points an attribute receives, the more important an attribute is. If an attribute is not at all important, assign it no points. If an attribute is twice as important as some other attribute, it should receive twice as many points.

Note: the figures below represent the mean points allocated to bottled beers by three segments of a target market.

Form

MEAN POINTS ALLOCATED				
	Attribute	**Segment I**	**Segment II**	**Segment III**
1	Bitterness	8	2	17
2	Hop flavours	2	4	20
3	Fragrance	3	9	19
4	Country where brewed	9	17	4
5	Price	53	5	7
6	High alcohol level	7	60	9
7	Aftertaste	5	0	15
8	Package design	13	3	9
	Sum	**100**	**100**	**100**

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Other Comparative Estimators

- Q-Sort Scaling developed to discriminate among a relatively large number of objects quickly and uses a rank order procedure in which objects are sorted into piles based on similarity with respect to some criterion.
- In magnitude estimation, numbers are assigned to objects such that ratios between the assigned numbers reflect ratios on the specified criterion

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Verbal Protocols

- A technique used to understand participants' cognitive responses or thought processes by having them think aloud while completing a task or making a decision.

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Non-Comparative Scales

Each object is evaluated independently of the others — also called monadic scales. Data are generally interval- or ratio-scaled. Two families: continuous rating scales and itemised rating scales.

Continuous Rating Scale

Participants place a mark on a continuous line to rate an object.

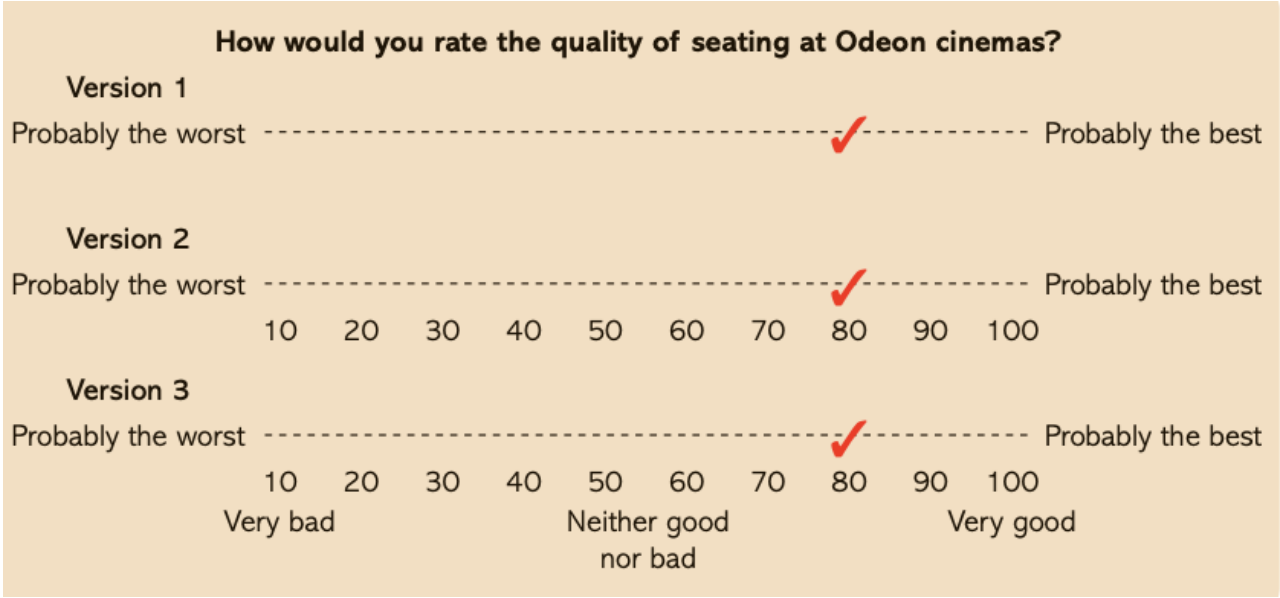
Easy to construct, but scoring can be cumbersome unless computerised.

Itemised Rating Scale

A number or brief description is given for each ordered category; participants select the category that best describes the object.

Common types: Likert, semantic differential, Stapel — see next slide.

Continuous Rating Scale



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Itemised Rating Scales Compared

Scale	Basic characteristic	Advantage	Disadvantage
Likert	Agreement, 1 (strongly disagree) to 5 (strongly agree)	Easy to construct, administer, understand	More time-consuming
Semantic differential	7-point scale with bipolar label pairs	Versatile	Debate on whether data are interval
Stapel	Unipolar, -5 to +5, no neutral point	Easy to construct; works by phone	Confusing and difficult to apply

Rule of thumb: 5–9 categories, generally balanced, with an odd number if a neutral response is meaningful.

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Likert Scale

Instructions
Listed below are different beliefs about the Odeon cinema. Please indicate how strongly you agree or disagree with each by using the following scale:

1 = Strongly disagree, 2 = Disagree, 3 = Neither agree nor disagree, 4 = Agree, 5 = Strongly agree

	1	2	3	4	5
1 I like to visit Odeon cinemas		✓			
2 The Odeon sells poor-quality food		✓			
3 The Odeon presents a wide variety of film genres			✓		
4 I do not like Odeon advertisements				✓	
5 The Odeon charges fair prices				✓	
6 Booking a seat at the Odeon is difficult	✓				
7 The acoustics at Odeon cinemas are excellent				✓	
8 Odeon staff serve their customers very well				✓	
9 The Odeon is a great place for families to enjoy films		✓			

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Semantic Differential Scale

Instructions
What does visiting an Odeon cinema mean to you? The following descriptive scales, bounded at each end by bipolar adjectives, summarise characteristics of a visit. Please mark X in the blank that best indicates what a visit to an Odeon cinema means to you.

Boring						X		Exciting
Special		X						Routine
Thrifty			X					Indulgent
Cosy		X						Uncomfortable
Peaceful						X		Noisy
Youthful		X						Mature

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The Stapel Scale

Instructions
Please evaluate how accurately each phrase describes a visit to an Odeon cinema. Select a positive number for the phrases you think describe a visit accurately. The more accurately you think the phrase describes a visit, the larger the positive number you should choose. You should select a minus number if you do not think the phrase accurately describes a visit. The less accurately you think it describes a visit, the larger the negative number you should choose. You can select any number from +5 for phrases you think are very accurate, to –5 for phrases you think are very inaccurate.

A visit to an Odeon cinema

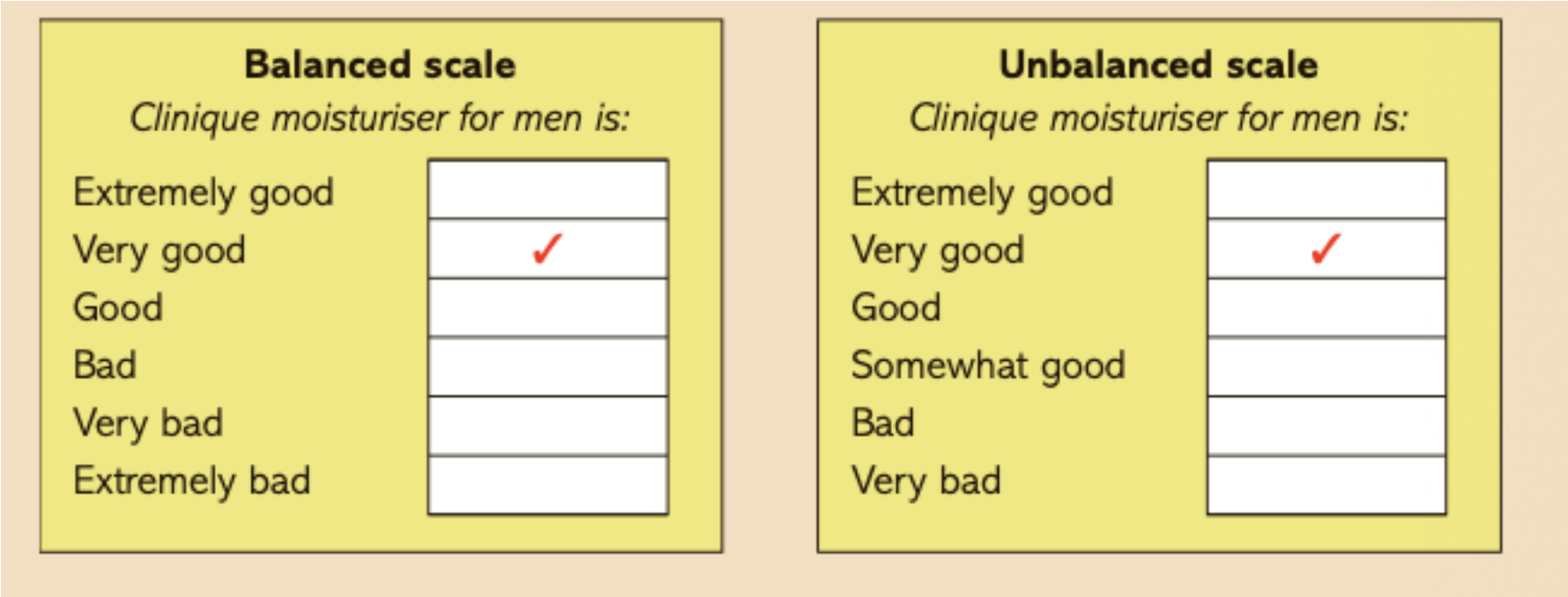
+5	+5
+4	+4
+3	+3
+2	+2 ✗
+1	+1
A special event	Full of energy
–1	–1
–2 ✗	–2
–3	–3
–4	–4
–5	–5

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Designing Itemised Rating Scales

1. Number of categories — typically five to nine
2. Balanced vs. unbalanced — balanced gives more objective data
3. Odd vs. even categories — odd allows a neutral response
4. Forced vs. unforced choice — unforced when “no opinion” is likely
5. Nature and degree of verbal description — label categories clearly
6. Physical form of the scale — test a few layouts and pick the best

Balanced vs Unbalanced Scale



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Common Marketing Scales

Construct	1	2	3	4	5
Attitude	Very bad	Bad	Neither bad nor good	Good	Very good
Importance	Not at all important	Not important	Neutral	Important	Very important
Satisfaction	Very dissatisfied	Dissatisfied	Neither dis- nor satisfied	Satisfied	Very satisfied
Purchase intent	Definitely will not buy	Probably will not buy	Might or might not buy	Probably will buy	Definitely will buy
Purchase frequency	Never	Rarely	Sometimes	Often	Ver

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Measurement Accuracy: the True Score Model

$$X_o = X_t + X_s + X_r$$

Observed score = True score + Systematic error + Random error

Potential sources of error

- Stable individual traits (intelligence, education)
- Transient personal factors (health, mood, fatigue)
- Situational factors (other people, noise, distractions)
- Sampling of scale items

- Lack of clarity in scale or instructions
- Mechanical factors (printing, layout)
- Administration differences (interviewers)
- Analysis differences (scoring, statistics)

Reliability

The extent to which measures are free from random error (X_r). If $X_r = 0$, the measure is perfectly reliable.

Test–retest

Same scale administered to the same people at two points in time; results are compared.

Alternative–forms

Two equivalent forms of the scale given to the same people at two points in time.

Split-half

Scale items are divided into two halves; the resulting half-scores are correlated.

Cronbach's alpha

Average of all possible split-half coefficients. Ranges 0–1; ≤0.6 is generally unsatisfactory.

Validity

The extent to which observed scale scores reflect true differences among objects, rather than systematic or random error.

Content validity

Subjective, systematic check that the scale covers the measurement task.

Criterion validity

Concurrent (measured at the same time) or predictive (forecasts a future outcome).

Construct validity

Convergent, discriminant and nomological validity — what the scale truly measures.

Reliability is necessary but not sufficient for validity: an unreliable measure cannot be valid, but a reliable one may still carry systematic error.

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